

# Francesco Emanuele Stradi

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## ABOUT ME

I am a postdoctoral researcher at Politecnico di Milano, working with Prof. Alberto Marchesi. I received my Ph.D. in computer science at Politecnico di Milano, where I was advised by Prof. Nicola Gatti. My current research focuses on the theoretical aspects of artificial intelligence, particularly online reinforcement learning (RL) and online learning. Specifically, I am interested in combining online (machine) learning theoretical tools with optimization techniques to build strategic agents able to act in complex online environments.

## EXPERIENCE

- **Politecnico di Milano** May 2026 - now  
Postdoctoral researcher  
◦ Working with Prof. Alberto Marchesi  
Milan, Italy
- **lastminute.com** June - September 2024  
Machine Learning Scientist, internship  
◦ In charge of developing online learning algorithms for dynamic pricing  
Chiasso, Switzerland
- **Polimi Artificial Intelligence Research and Innovation Center (Airic)** August - October 2022  
Research Scientist, internship  
◦ In charge of selecting State-Of-The-Art artificial intelligence techniques and applying them in real world business oriented scenarios  
Milan, Italy

## EDUCATION

- **Politecnico di Milano** 2022 - 2026  
PhD in Computer Science  
◦ Advised by Prof. Nicola Gatti  
◦ Conferred *cum laude*  
Milan, Italy
- **Columbia University** February - April 2025  
Visiting Scholar  
◦ Working with Prof. Christian Kroer at IEOR Department  
New York City, USA
- **Politecnico di Milano** 2019 - 2022  
Msc in Computer Science and Engineering  
◦ Grade: 110 cum laude/110  
Milan, Italy
- **Politecnico di Milano** 2016 - 2019  
Bsc in Computer Engineering  
Milan, Italy

## PUBLICATIONS

C=CONFERENCE, J=JOURNAL, W=WORKSHOP, T=THESIS

- [C.1] **Francesco Emanuele Stradi**, Kalana Kalupahana, Matteo Castiglioni, Alberto Marchesi and Nicola Gatti (2026). **Truly Adapting to Adversarial Constraints in Constrained MABs**. In *The 39th Annual Conference on Learning Theory [COLT 2026]*.
- [C.2] **Francesco Emanuele Stradi**, Matteo Castiglioni, Alberto Marchesi, Nicola Gatti and Christian Kroer (2025). **No-Regret Learning Under Adversarial Resource Constraints: A Spending Plan Is All You Need!**. In *The 39th Conference on Neural Information Processing Systems [NeurIPS 2025]*.
- [C.3] Gianmarco Genalti\*, **Francesco Emanuele Stradi**\*, Matteo Castiglioni, Alberto Marchesi and Nicola Gatti (2025). **Data-Dependent Regret Bounds for Constrained MABs**. In *The 39th Conference on Neural Information Processing Systems [NeurIPS 2025]*.
- [C.4] **Francesco Emanuele Stradi**, Anna Lunghi, Matteo Castiglioni, Alberto Marchesi and Nicola Gatti (2025). **Taming Adversarial Constraints in CMDPs**. In *The 39th Conference on Neural Information Processing Systems [NeurIPS 2025]*.
- [C.5] Francesco Bacchiocchi\*, **Francesco Emanuele Stradi**\*, Matteo Castiglioni, Alberto Marchesi and Nicola Gatti (2025). **Markov Persuasion Processes: Learning to Persuade From Scratch**. In *The 39th Conference on Neural Information Processing Systems [NeurIPS 2025]*.
- [C.6] **Francesco Emanuele Stradi**, Matteo Castiglioni, Alberto Marchesi and Nicola Gatti (2025). **Learning Adversarial MDPs with Stochastic Hard Constraints**. In *The 42nd International Conference on Machine Learning [ICML 2025]*.

- [C.7] **Francesco Emanuele Stradi**, Anna Lunghi, Matteo Castiglioni, Alberto Marchesi and Nicola Gatti (2025). **Policy Optimization for CMDPs with Bandit Feedback: Learning Stochastic and Adversarial Constraints**. In *The 42nd International Conference on Machine Learning [ICML 2025]*.
- [C.8] **Francesco Emanuele Stradi**, Matteo Castiglioni, Alberto Marchesi and Nicola Gatti (2025). **Optimal Strong Regret and Violation in Constrained MDPs via Policy Optimization**. In *The 13th International Conference on Learning Representations [ICLR 2025]*.
- [C.9] Davide Maran\*, Francesco Bacchiocchi\*, **Francesco Emanuele Stradi\***, Matteo Castiglioni, Nicola Gatti and Marcello Restelli (2024). **Bandits with Ranking Feedback**. In *The 38th Conference on Neural Information Processing Systems [NeurIPS 2024]*.
- [C.10] **Francesco Emanuele Stradi**, Jacopo Germano, Gianmarco Genalti, Matteo Castiglioni, Alberto Marchesi and Nicola Gatti (2024). **Online Learning in CMDPs: Handling Stochastic and Adversarial Constraints**. In *The 41st International Conference on Machine Learning [ICML 2024]*.
- [C.11] Francesco Bacchiocchi\*, **Francesco Emanuele Stradi\***, Matteo Papini, Alberto Maria Metelli and Nicola Gatti (2024). **Online Learning with Off-Policy Feedback in Adversarial MDPs**. In *The 33rd International Joint Conference on Artificial Intelligence [IJCAI 2024]*.
- [C.12] Davide Maran\*, Pierricardo Olivieri\*, **Francesco Emanuele Stradi\***, Nicola Gatti and Marcello Restelli (2024). **Online Markov Decision Processes Configuration with Continuous Decision Space**. In *The 38th AAAI Conference on Artificial Intelligence [AAAI 2024]*.
- [J.1] **Francesco Emanuele Stradi**, Anna Lunghi, Matteo Castiglioni, Alberto Marchesi and Nicola Gatti (2026). **Policy Optimization for CMDPs with Bandit Feedback: Best-of-Both-Worlds and Beyond**. In *Artificial Intelligence [AIJ 2026]*.
- [W.1] Gianmarco Genalti\*, **Francesco Emanuele Stradi\***, Matteo Castiglioni, Alberto Marchesi and Nicola Gatti (2025). **Data-Dependent Regret Bounds for MABs with Constraints**. In *NeurIPS 2025 Workshop: Aligning Reinforcement Learning Experimentalists and Theorists [ARLET 2025]*.
- [W.2] **Francesco Emanuele Stradi**, Anna Lunghi, Matteo Castiglioni, Alberto Marchesi and Nicola Gatti (2025). **Policy Optimization in CMDPs with Bandit Feedback: Learning with Stochastic and Adversarial Constraints**. In *NeurIPS 2025 Workshop: Aligning Reinforcement Learning Experimentalists and Theorists [ARLET 2025]*.
- [W.3] **Francesco Emanuele Stradi**, Anna Lunghi, Matteo Castiglioni, Alberto Marchesi and Nicola Gatti (2025). **Taming Adversarial Constraints in Constrained MDPs**. In *The 18th European Workshop on Reinforcement Learning [EWRL 2025]*.
- [W.4] **Francesco Emanuele Stradi**, Anna Lunghi, Matteo Castiglioni, Alberto Marchesi and Nicola Gatti (2025). **Policy Optimization for CMDPs with Bandit Feedback: Learning Stochastic and Adversarial Constraints**. In *The 18th European Workshop on Reinforcement Learning [EWRL 2025]*.
- [W.5] **Francesco Emanuele Stradi**, Matteo Castiglioni, Alberto Marchesi and Nicola Gatti (2024). **Online learning in CMDPs with adversarial losses and stochastic hard constraints**. In *The 17th European Workshop on Reinforcement Learning [EWRL 2024]*.
- [W.6] Francesco Bacchiocchi\*, **Francesco Emanuele Stradi\***, Matteo Castiglioni, Alberto Marchesi and Nicola Gatti (2024). **Markov Persuasion Processes: How to Persuade Multiple Agents From Scratch**. In *ICML 2024 Workshop: Aligning Reinforcement Learning Experimentalists and Theorists [ARLET 2024]*.
- [W.7] Francesco Bacchiocchi\*, **Francesco Emanuele Stradi\***, Matteo Papini, Alberto Maria Metelli and Nicola Gatti (2023). **Online Adversarial MDPs with Off-Policy Feedback and Known Transitions**. In *The 16th European Workshop on Reinforcement Learning [EWRL 2023]*.
- [W.8] Davide Maran\*, Pierricardo Olivieri\*, **Francesco Emanuele Stradi\***, Nicola Gatti and Marcello Restelli. (2023). **Online Configuration in Continuous Decision Space**. In *The 16th European Workshop on Reinforcement Learning [EWRL 2023]*.
- [T] **Francesco Emanuele Stradi** (2026). **Regret Minimization in Constrained Markov Decision Processes**. External Reviewers: Prof. Haipeng Luo, Matteo Pirodda, PhD.

\* stands for equal contribution

## TEACHING

- **Probabilità e Statistica Matematica, TA**  
Politecnico di Milano
  - Bsc in Management Engineering
- **Statistica, TA**  
Politecnico di Milano
  - Bsc in Mechanical Engineering

Spring 2024

Spring 2024

## SERVICE

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- **Reviewing:** ICLR (2025, 2026), ICML (2025, 2026 **Silver Reviewer**), NeurIPS (2024, 2025, 2026), Mathematics of Operations Research (2025), Transactions on Machine Learning Research (2026)
- **Program Committee:** EC (2026)

## PROGRAMMING

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- **Programming Languages:** C, Java, Python, SQL, Ampl

## ADDITIONAL INFORMATION

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- **Languages:** Italian (Native), English (Proficiency level)

## REFERENCES

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**Prof. Nicola Gatti**  
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Politecnico di Milano  
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**Prof. Christian Kroer**  
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